



THE UNIVERSITY OF
SYDNEY

Room Number _____
Seat Number _____
Student Number

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ANONYMOUSLY MARKED

(Please do not write your name on this exam paper)

CONFIDENTIAL EXAM PAPER

This paper is not to be removed from the exam venue

Mathematics and Statistics

EXAMINATION

Semester 1 - Main, 2018

**MATH1921/MATH1931 Calculus Of One Variable
(Advanced)/(SSP)**

For Examiner Use Only

EXAM WRITING TIME: 1.5 hours
READING TIME: 10 minutes

EXAM CONDITIONS:

This is a CLOSED book examination - no material permitted

During reading time - writing is not permitted at all

MATERIALS PERMITTED IN THE EXAM VENUE:

(No electronic aids are permitted e.g. laptops, phones)

Calculator - non-programmable

MATERIALS TO BE SUPPLIED TO STUDENTS:

Dept-specific answer material - provided by dept

INSTRUCTIONS TO STUDENTS:

This examination has two sections: Multiple Choice and Extended Answer.

The Multiple Choice Section is worth 30% of the total examination; there are 15 questions; the questions are of equal value; all questions may be attempted. Answers to the Multiple Choice questions must be entered on the Multiple Choice Answer Sheet.

The Extended Answer Section is worth 70% of the total examination; there are 4 questions; the questions are of equal value; all questions may be attempted; working must be shown.

A list of standard derivatives and integrals is provided at the back of the booklet.

Q	Mark
1	
2	
3	
4	

Please tick the box to confirm that your examination paper is complete. ☐

Extended Answer Section

There are **four** questions in this section, each with a number of parts.
Write the answers in the booklets provided.

1. (a) (i) Write the complex number $z = -1 - \sqrt{3}i$ in modulus–argument form.

Solution: The modulus is $|z| = \sqrt{(-1)^2 + (-\sqrt{3})^2} = 2$. Taking into account the quadrant, the argument is $\arg(z) = -\frac{2\pi}{3}$.

- (ii) Write $3e^{\frac{\pi}{3}i}e^{\frac{\pi}{6}i}$ in Cartesian form.

Solution: We have

$$3e^{\frac{\pi}{3}i}e^{\frac{\pi}{6}i} = 3e^{\frac{\pi}{2}i} = 3\left(\cos\left(\frac{\pi}{2}\right) + i\sin\left(\frac{\pi}{2}\right)\right) = 3i.$$

- (b) Determine whether or not the function $f: \mathbb{R} \rightarrow \mathbb{C}$ given by $f(t) = te^{it}$ is injective.

Solution: We test whether there exist distinct $s, t \in \mathbb{R}$ such that $f(s) = f(t)$. Taking the modulus we have

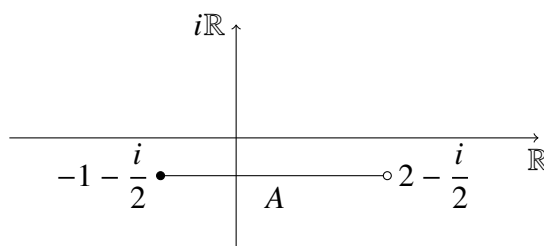
$$|t| = |te^{it}| = |se^{is}| = |s|.$$

Hence $s = \pm t$. Since we look for distinct values we have $s = -t$. Substitution gives $te^{it} = f(t) = f(-t) = -te^{-it}$, which implies that $e^{it} = -e^{-it}$ or $e^{2it} = -1$. This is satisfied if $2t = \pi$ or $t = \pi/2$. Hence we have $f(\pi/2) = f(-\pi/2)$, showing that f is not injective.

- (c) Let A be the line $\operatorname{Im} z = -\frac{1}{2}$ in the complex plane with $-1 \leq \operatorname{Re} z < 2$.

- (i) Sketch the set A on the complex plane.

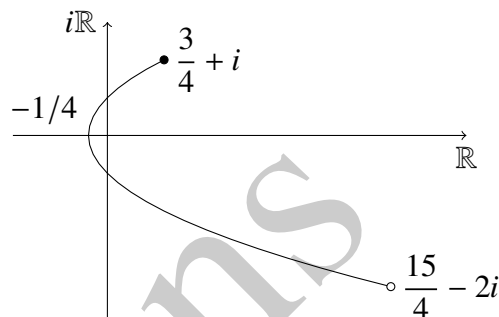
Solution:



- (ii) Sketch the image of A under the map $z \mapsto z^2$ on the complex plane.
Carefully label the x -intercepts and the endpoints.

Solution: Every point on A has the form $z = t - \frac{i}{2}$ with $-1 \leq t < 2$. Squaring gives $z^2 = t^2 - it - \frac{1}{4}$. The real part is $u = t^2 - \frac{1}{4}$ and the imaginary part is $v = -t$.

Hence the shape is a sideways parabola with vertex at $x = -\frac{1}{4}$. The endpoints are $\frac{3}{4} + i$ (included) and $\frac{15}{4} - 2i$ (excluded). The graph is shown below.



2. (a) Find the 4th order Taylor polynomial of $f(x) = \sqrt{1 + 4x^2}$ centred at $x = 0$.

Solution: If we set $g(t) = \sqrt{1 + t}$, then $f(x) = g((2x)^2)$. Hence we only need to find the 2nd order Taylor polynomial of g . We have

$$\begin{aligned} g(t) &= \sqrt{1+t} & g(0) &= 1 \\ g'(t) &= \left(\frac{1}{2}\right) \frac{1}{\sqrt{1+t}} & g'(0) &= \frac{1}{2} \\ g''(t) &= \left(-\frac{1}{2}\right) \left(\frac{1}{2}\right) \frac{1}{(\sqrt{1+t})^3} & g''(0) &= -\frac{1}{4}. \end{aligned}$$

The corresponding Taylor polynomial is

$$1 + \frac{t}{2} - \frac{t^2}{8}.$$

Hence the 4th order Taylor polynomial of f is

$$T_4(x) = 1 + \frac{(2x)^2}{2} - \frac{(2x)^4}{8} = 1 + 2x^2 - 2x^4.$$

- (b) Is it possible to construct a function $f: [1, 3] \rightarrow \mathbb{R}$ that is continuous on $[1, 3]$, differentiable on $(1, 3)$ and for which $f(1) = -1$, $f(3) = 4$ and $f'(x) \leq 2$ for all $x \in (1, 3)$? Provide some explanation for your answer.

Solution: By the mean value theorem there exists $c \in (1, 3)$ such that

$$f'(c) = \frac{f(3) - f(1)}{3 - 1} = \frac{5 - (-1)}{2} = \frac{6}{2} = 3 > 2.$$

As $f'(c) \leq 2$ by assumption it is not possible to find such a function.

(c) Compute $\lim_{x \rightarrow \infty} \left(\frac{x^2}{x+3} - \frac{x^2}{x-3} \right)$.

Solution: If we make common denominators we can bring the expression in a form to apply the limit laws:

$$\frac{x^2}{x+3} - \frac{x^2}{x-3} = -\frac{6x^2}{(x-3)(x+3)} = \frac{-6x^2}{x^2-9} = \frac{-6}{1-\frac{9}{x^2}} \xrightarrow{x \rightarrow \infty} -6.$$

Alternatively use l'Hôpital's rule by first rewriting the expression in the form

$$\frac{x^2}{x+3} - \frac{x^2}{x-3} = \frac{-6x^2}{x^2-9}$$

as before and observing that the last expression is of the form $\frac{-\infty}{\infty}$. Then differentiate numerator and denominator

$$\frac{(-6x^2)'}{(x^2-9)'} = \frac{-12x}{2x} = -6 \xrightarrow{x \rightarrow \infty} -6.$$

3. (a) Compute $\frac{d}{dx} \int_0^{\sqrt{x}} \frac{t^2}{1+t^4} dt$ given that $x > 0$.

Solution: Using the fundamental theorem of calculus and the chain rule we have

$$\frac{d}{dx} \int_0^{\sqrt{x}} \frac{t^2}{1+t^4} dt = \frac{(\sqrt{x})^2}{1+(\sqrt{x})^4} \frac{d\sqrt{x}}{dx} = \frac{x}{1+x^2} \frac{1}{2\sqrt{x}}.$$

- (b) Compute the definite integral $\int_0^1 \frac{x-2}{(x+2)(x+3)} dx$.

Solution: Using partial fractions we have

$$\frac{x-2}{(x+2)(x+3)} = \frac{5}{x+3} - \frac{4}{x+2}$$

Integration yields

$$\begin{aligned} \int_0^1 \frac{5}{x+3} - \frac{4}{x+2} dx &= [-4 \log(x+2) + 5 \log(x+3)]_0^1 \\ &= -9 \log(3) + 4 \log(2) + 5 \log(4). \end{aligned}$$

- (c) (i) Let $A > 0$ be a constant and let $n, m \in \mathbb{N}$ so that $n > m$. Show that

$$\frac{A^n}{n!} \leq \frac{A^m}{m!} \left(\frac{A}{m+1} \right)^{n-m}.$$

Solution: We can write

$$\frac{A^n}{n!} = \frac{A^m}{m!} \underbrace{\frac{A^{n-m}}{(m+1)(m+2)\cdots n}}_{n-m \text{ terms}} \leq \frac{A^m}{m!} \frac{A^{n-m}}{(m+1)^{n-m}} = \frac{A^m}{m!} \left(\frac{A}{m+1} \right)^{n-m},$$

where we used for the inequality that $(m+1)(m+2)\cdots n \geq (m+1)^{n-m}$.

- (ii) Hence, or otherwise, show that $\frac{A^n}{n!} \rightarrow 0$ as $n \rightarrow \infty$ for every constant $A > 0$.

Solution: Given $A > 0$ we choose $m \in \mathbb{N}$ so that $m+1 > A$. Then

$$0 < \frac{A}{m+1} < 1.$$

Hence, $\left(\frac{A}{m+1} \right)^{n-m} \rightarrow 0$ as $n \rightarrow \infty$. As

$$0 \leq \frac{A^n}{n!} \leq \frac{A^m}{m!} \left(\frac{A}{m+1} \right)^{n-m}$$

the squeeze law implies that

$$\frac{A^n}{n!} \xrightarrow{n \rightarrow \infty} 0.$$

- (iii) By estimating the remainder term in the n -th order Taylor polynomial of $f(x) = e^x$, prove that

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

for all $x \in \mathbb{R}$.

Solution: The Lagrange form of the remainder is

$$R_n(x) = \frac{d}{dx} e^c \Big|_{x=c} \frac{x^{n+1}}{(n+1)!} = e^c \frac{x^{n+1}}{(n+1)!}$$

for some c between 0 and x . As e^x is increasing we have

$$|R_n(x)| \leq e^{|x|} \frac{|x|^{n+1}}{(n+1)!}$$

which converges to zero as $n \rightarrow \infty$ by part (ii). Hence the Taylor series converges and represents e^x .

4. (a) Compute the indefinite integral $\int \tan^3 x \, dx$.

Solution: Using that $\sec^2 x = 1 + \tan^2 x$ we can write

$$\tan^3 x = (\sec^2 x - 1) \tan x = \sec^2 x \tan x - \tan x = \sec^2 x \tan x - \frac{\sin x}{\cos x}.$$

Since $(\tan x)' = \sec^2 x$ and $(\cos x)' = -\sin x$ we have

$$\int \tan^3 x \, dx = \int \sec^2 x \tan x \, dx + \int \frac{-\sin x}{\cos x} \, dx = \frac{1}{2} \tan^2 x + \ln |\cos x| + C.$$

(b) Determine whether or not the improper integral $\int_0^\infty \frac{\sin x}{x^{3/2}} dx$ exists.

Solution: We use the comparison theorem to do so. First note that $\frac{|\sin x|}{|x|} \leq 1$ for all $x > 0$ and hence

$$\left| \frac{\sin x}{x^{3/2}} \right| \leq \frac{1}{\sqrt{x}}$$

for all $x \in (0, 1]$. Since

$$\int_0^1 \frac{1}{\sqrt{x}} dx < \infty$$

the integral

$$\int_0^1 \frac{\sin x}{x^{3/2}} dx$$

exists by the comparison theorem. Moreover, as $|\sin x| \leq 1$ we have

$$\left| \frac{\sin x}{x^{3/2}} \right| \leq \frac{1}{x^{3/2}}.$$

for all $x \in [1, \infty)$. Since

$$\int_1^\infty \frac{1}{x^{3/2}} dx < \infty$$

the integral

$$\int_1^\infty \frac{\sin x}{x^{3/2}} dx$$

exists by the comparison theorem. Both together show that the integral exists on $(0, \infty)$.

(c) Assume that $[a, b]$ is a closed bounded interval and that $f: [a, b] \rightarrow \mathbb{R}$ is continuous and not constant. Use the intermediate value theorem and the extreme value theorem to prove that the image $\text{im}(f) := \{f(x) \mid x \in [a, b]\}$ of f is a closed and bounded interval.

Solution: According to the extreme value theorem, f has a maximum and a minimum, that is, there exist $x_1, x_2 \in [a, b]$ such that

$$m := f(x_1) \leq f(x) \leq f(x_2) = M$$

for all $x \in [a, b]$. Hence $\text{im}(f) \subseteq [m, M]$.

Since f is non-constant either $x_1 < x_2$ or $x_2 < x_1$. As f is continuous on $[x_1, x_2]$ or $[x_2, x_1]$ respectively, the intermediate value theorem implies that for every k between m and M there exists c between x_1 and x_2 so that $f(c) = k$. Hence $[m, M] \subseteq \text{im}(f)$.

Both inclusions together imply that $\text{im}(f) = [m, M]$ is closed and bounded.

Table of Standard Integrals

$$1. \int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$9. \int \sec^2 x dx = \tan x + C$$

$$2. \int \frac{dx}{x} = \ln |x| + C$$

$$10. \int \operatorname{cosec}^2 x dx = -\cot x + C$$

$$3. \int e^x dx = e^x + C$$

$$11. \int \sec x dx = \ln |\sec x + \tan x| + C$$

$$4. \int \sin x dx = -\cos x + C$$

$$12. \int \operatorname{cosec} x dx = \ln |\operatorname{cosec} x - \cot x| + C$$

$$5. \int \cos x dx = \sin x + C$$

$$13. \int \sinh x dx = \cosh x + C$$

$$6. \int \tan x dx = -\ln |\cos x| + C$$

$$14. \int \cosh x dx = \sinh x + C$$

$$7. \int \cot x dx = \ln |\sin x| + C$$

$$15. \int \tanh x dx = \ln \cosh x + C$$

$$8. \int \frac{dx}{a^2 + x^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$$

$$16. \int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \left(\frac{x}{a} \right) + C \quad (|x| < a)$$

$$17. \int \frac{dx}{\sqrt{x^2 + a^2}} = \sinh^{-1} \left(\frac{x}{a} \right) + C = \ln \left(x + \sqrt{x^2 + a^2} \right) + C'$$

$$18. \int \frac{dx}{\sqrt{x^2 - a^2}} = \cosh^{-1} \left(\frac{x}{a} \right) + C = \ln \left(x + \sqrt{x^2 - a^2} \right) + C' \quad (x > a)$$

$$\text{Linearity: } \int (\lambda f(x) + \mu g(x)) dx = \lambda \int f(x) dx + \mu \int g(x) dx$$

$$\text{Integration by substitution: } \int f(u(x)) \frac{du}{dx} dx = \int f(u) du$$

$$\text{Integration by parts: } \int f(x) g'(x) dx = f(x) g(x) - \int f'(x) g(x) dx$$

End of Extended Answer Section

END OF EXAMINATION